MATH 566 : Lecture 23 (11/10/2026)

Today:
* MCF — assumptions
* residual network for MCF
* MCF optimality conditions

The Min-Cost Flow (MCF) Problem

Recall SP optimality conditions: $d(j) \le d(i) + c_{ij} \ \forall (i,j) \in A$.

Max flow optimality conditions: No augmenting paths in $G(\bar{x})$.

In MCF, we work with costs, capacities, and supplies/demand. We use $C = \max_{(i,j)\in A} \{|c_{ij}|\}$ and $U = \max_{(i,j)\in A} u_{ij}$ in our discussion.

Optimization model (Linear Program):

$$\min \sum_{(i,j)\in A} c_{ij} x_{ij} \qquad \text{(total cost)}$$

$$\text{s.t.} \quad \sum x_{ij} - \sum x_{ji} = b(i) \ \forall i \in N \qquad \text{(flow balance)}$$

outflow; inflow; supply/demand

$$0 \le x_{ij} \le u_{ij} \ \forall (i,j) \in A \qquad \text{(bounds)}$$

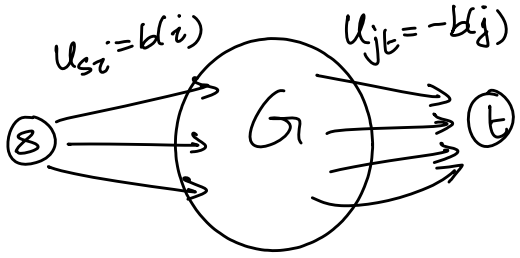
Again, we will **not** directly solve MCF problems as linear programs. The focus will be on efficient algorithms.

AMO describes several applications modeled as MCF — we will not discuss them here.

Assumptions

1. $l_{ij} = 0$ → else remove nonzero lower bounds (network transformations)
2. All data is integral $(c_{ij}, u_{ij}, b(i))$ for complexity analysis purposes
3. The network is directed.
4. $\sum_{i \in N} b(i) = 0$ (total supply = total demand)
   → else, MCF instance is not feasible
5. The MCF problem has a feasible solution.

Recall: Given an MCF problem instance, we can check if it has a feasible flow by solving a max-flow problem.



Add s, t, (s,i) with $u_{si} = b(i)$ for i s.t. $b(i) > 0$ (supply nodes), and (j,t) with $u_{jt} = -b(j)$ for j s.t. $b(j) < 0$ (demand nodes).

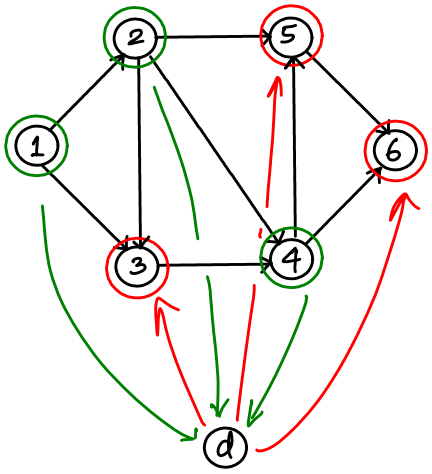
If an s-t max flow saturates all these extra arcs, the original MCF instance has a feasible solution.

Another approach to handle feasibility:

(i) supply node, $b(i) > 0$

(j) demand node, $b(j) < 0$

G



} artificial arcs

d: dummy node (transshipment node) $b(d) = 0$.

Add arcs (i,d) $\forall i \in N$ with $b(i) > 0$, and (d,j) $\forall j \in N$ with $b(j) < 0$, with $c_{id} = c_{dj} = +\infty$ and $u_{id} = u_{dj} = +\infty$. Solve MCF on this modified problem. If the optimal solution has any $x_{id} > 0$ or $x_{dj} > 0$, then the original MCF has no feasible solution. If not, x_{ij} $\forall (i,j) \in A$ (in original network G) is an optimal solution to the MCF problem.

This option is motivated by the use of artificial variables in linear programming (the big-M simplex method).

Assumptions (for MCF, continued)

6. There is an uncapacitated directed path between every pair of nodes. Can add extra arc (i,j) with $u_{ij} = \infty$, $c_{ij} = +\infty$ if needed.

7. $c_{ij} \geq 0 \quad \forall (i,j) \in A$ (else, we can use arc reversal)

We need $u_{ij} < \infty$ for this transformation.

If $u_{ij} = \infty$, use $u_{ij} = B > \sum_{\substack{u_{ij} \text{ is} \\ \text{finite}}} u_{ij} + \sum_{b(i)>0} b(i)$.

finite, but large enough to act as $+\infty$.

8. If some $u_{ij} = +\infty$, we assume there is no negative cost cycle of infinite capacity.

If there is such a cycle, the problem is unbounded. Else, we could replace $u_{ij} = \infty$ with B as shown above.

Notice that we get Assumption 8 when Assumption 7 is satisfied. At the same time, it is helpful to list them both separately.

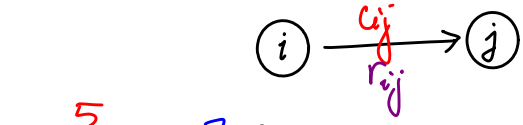
Residual Network for Min-Cost Flow

Recall: residual network for max flow: $r_{ij} = u_{ij} - x_{ij} + x_{ji}$. But in MCF c_{ij} may not be same as c_{ji}, so we cannot combine residual capacities usually.

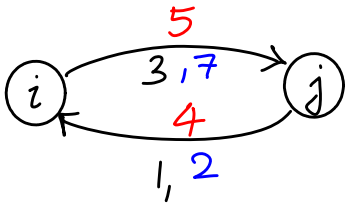
$G(\bar{x})$ for MCF: for arc (i,j) we get

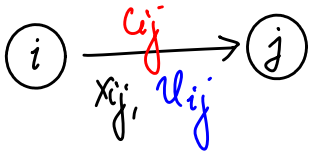
$r_{ij} = u_{ij} - x_{ij}$ with cost c_{ij}, and
$r_{ji} = x_{ij}$ with cost $-c_{ij}$ (i.e., $c_{ji} = -c_{ij}$ here).

Of course, only those arcs (i,j) with $r_{ij} > 0$ appear in $G(\bar{x})$.

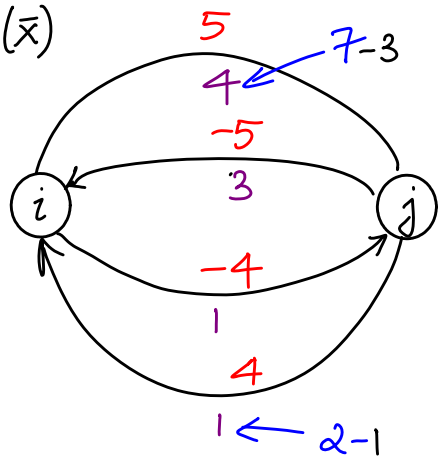


G:





$G(\bar{x})$



Recall that in max flow, we could combine the residual capacities arising from different arcs. In particular, we had $r_{ij} = (u_{ij} - x_{ij}) + x_{ji}$. But in MCF, the costs are different, and hence we cannot combine.

Reduced Costs

Recall, SP optimality conditions: $d(j) \le d(i) + c_{ij} \quad \forall (i,j) \in G_1$.

Equivalently, $c_{ij}^{d} = c_{ij} + d(i) - d(j) \ge 0 \quad \forall (i,j) \in G_1$.

Def For a set of node potentials $\pi(i)$, $i \in N$, $c_{ij}^{\pi} = c_{ij} - \pi(i) + \pi(j)$ is the **reduced cost** of (i,j) **with respect to** π.

In the SP case, we take $\pi(i) = -d(i)$.

Optimality Conditions for MCF

We extend the SP optimality conditions to $G_1(\bar{x})$ for MCF to define MCF optimality conditions. We then devise algorithms for MCF that check for these optimality conditions repeatedly, and modify $\bar{x}$ and $G_1(\bar{x})$ to correct any violations.

We present three different optimality conditions. Naturally, they are equivalent. The first two are specified on $G_1(\bar{x})$, while the third is specified on G_1 (original network).

1. Negative Cycle Optimality Conditions

AMO Theorem 9.1 A feasible flow $\bar{x}$ is an optimal solution to the MCF problem iff there is no negative-cost cycle in $G_1(\bar{x})$.

Proof

($\Rightarrow$) If there is a negative cycle $G_f(\bar{x})$, then augment flow along it to reduce total cost. Hence $\bar{x}$ is not an optimal flow. Hence by the contrapositive, if $\bar{x}$ is optimal, $G_f(\bar{x})$ has no negative cycle.

($\Leftarrow$) Let $\bar{x}$ be a feasible flow, and $G_f(\bar{x})$ has no negative cycle. We want to show $\bar{x}$ is optimal.

Assume $\bar{x}^*$ is an optimal flow, and $\bar{x}^* \neq \bar{x}$. Then we can decompose the difference $\bar{x}^* - \bar{x}$ into at most $2m$ cycles. ($G_f(\bar{x})$ has $\leq 2m$ arcs)

The sum of the costs of all these cycles is $\bar{c}^T\bar{x}^* - \bar{c}^T\bar{x}$. (see below for why)

Since $G_f(\bar{x})$ has no negative cycles, it must hold that $\bar{c}^T\bar{x}^* - \bar{c}^T\bar{x} \geq 0 \Rightarrow \bar{c}^T\bar{x}^* \geq \bar{c}^T\bar{x}$. But $\bar{x}^*$ is an optimal solution, i.e., $\bar{c}^T\bar{x}^* \leq \bar{c}^T\bar{x}$. So, $\bar{c}^T\bar{x} = \bar{c}^T\bar{x}^*$.

Hence $\bar{x}$ is also an optimal solution. □

Note: Here $\bar{c}^T\bar{x} = \sum_{(i,j)\in A} c_{ij}x_{ij}$, with $\bar{x} = \begin{bmatrix} 1 \\ 2 \\ \vdots \\ m \end{bmatrix} \begin{bmatrix} x_{ij} \end{bmatrix}$, the vector of flows x_{ij}.

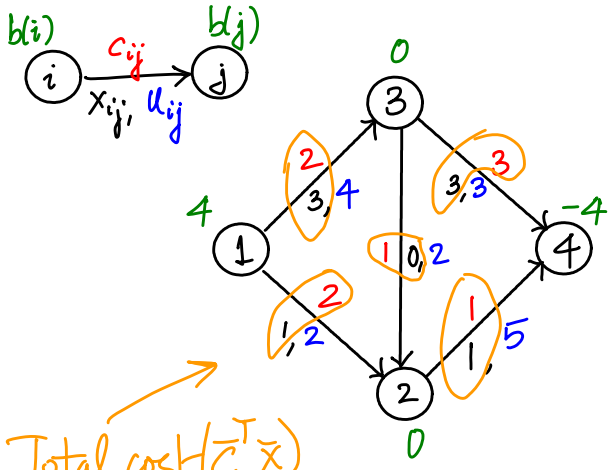
Also, since both $\bar{x}^*$ and $\bar{x}$ are feasible flows, their difference $\bar{x}^* - \bar{x}$ can be decomposed into only cycle flows. Due to the flow balance constraints, and (since $b(i)$ is same for both $\bar{x}^*$ and $\bar{x}$), we cannot get any path flows.

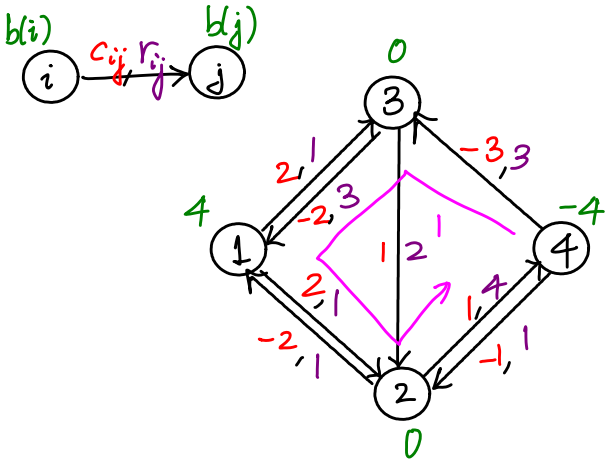
Negative Cycle Canceling Algorithm

* Start with a feasible flow $\bar{x}$.
* Use FIFO label correcting algorithm to identify a negative cycle in $G(\bar{x})$.
* augment, update $\bar{x}$, $G(\bar{x})$; repeat.

Example

G



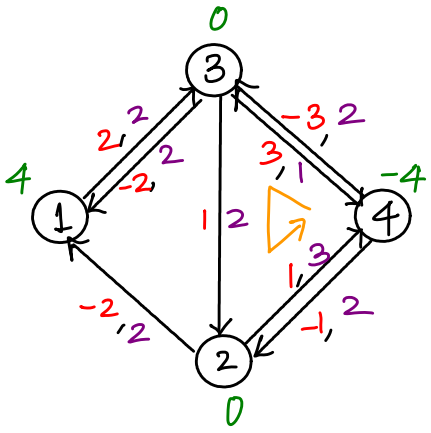


$W_1 = 1\text{-}2\text{-}4\text{-}3\text{-}1$

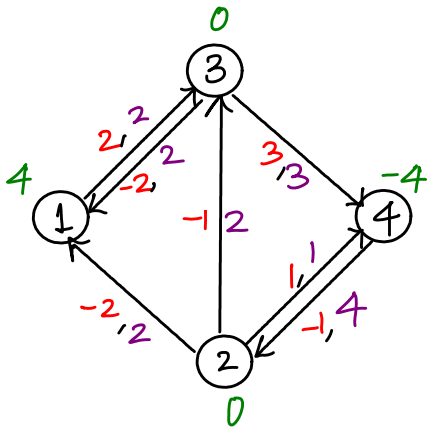
$c(W_1) = -2$

$\delta(W_1) = 1 \quad (r_{12} = 1)$

Augment $\delta = 1$ along W_1.



augment $\delta(W_2) = 2$



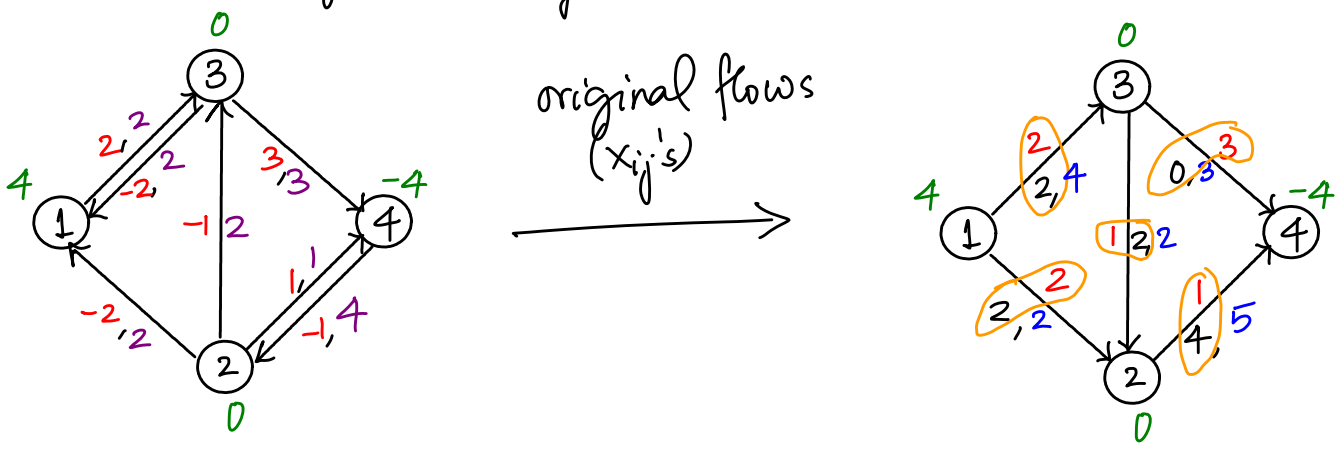
$W_2 = 2\text{-}4\text{-}3\text{-}2$

$c(W_2) = -1, \quad \delta(W_2) = 2$

r_{32} and r_{43}.

No more negative cycles, so flow is optimum.

We recover the original flows x_{ij}, similar to how we did it for max flow:



$\bar{c}^T\bar{x} = 14$ here

originally, $\bar{c}^T\bar{x} = 18$.

Finiteness and Complexity

AMO Theorem 9.10 If u_{ij} and $b(i)$ are all integers, the negative cycle canceling algorithm maintains an integral flow (solution) in each iteration.

Proof Initial feasible flow can be found using a max flow, which gives an integral flow. In each iteration, the bottleneck capacity is integral.

Theorem The negative cycle canceling algorithm is finite if all data is finite and integral. (finite → terminates after a finite # iterations)

Proof We use $-mCU$ as a lower bound and mCU as an upper bound for the total cost. Hence, the maximum change in total cost is $2mCU$.

In each iteration, total cost is decreased by at least 1. Hence, the algorithm terminates in at most $2mCU$ augmentations.

FIFO label correcting algorithm in each step takes $O(mn)$ time. Hence the overall time complexity of the negative cycle canceling algorithm is $O(m^2nCU)$. □